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Supernova Remnants and Stellar Winds

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ABSTRACT

We will look into major application of shock dynamics in astrophysical sites: supernova remnants (SNRs) and stellar winds. I mostly follow instructions given in Kwok (2007), with small pieces from Dopita & Sutherland (2003), Draine (2011), and Tielens (2005). Vink (2012) gives a relatively recent review of X-ray properties of SNR. The basics of stellar winds is summarized in Lamers & Cassinelli (1999). I also borrowed some of Christopher Hirata's lecture notes (winter 2011).

1. EVOLUTION OF A SN REMNANT IN A UNIFORM ISM

The evolution of Supernova Remnants (SNRs) is usually divided in four phases:

1. the ejecta dominated phase, in which the mass of the supernova ejecta, $M_{\rm ej}$, dominates over the swept-up mas $M_{\rm swept}$
2. the Sedov-Taylor phase, for which $M_{\rm swept} > M_{\rm ej}$, but for which radiative losses are not energetically important
3. the pressure-driven, or "snow-plow" phase, in which radiative cooling has become energetically important; the evolution of the shock radius is now best described by using momentum conservation
4. the merging phase, in which the shock velocity and temperature behind the shock become comparable to, respectively, the turbulent velocity and temperature of the interstellar medium.

For a list of SNR see https://en.wikipedia.org/wiki/List_of_supernova_remnants.

Although these discrete phases provide a useful framework to think of SNRs, it should be kept in mind that it is an oversimplification, and the phase of an individual SNR is not always that easily labeled.

1.1. *Setup*

The expanding mass of ejecta creates a shock that travels outwards through the ISM (**blast wave**), which accelerates ISM material away from the supernova. Similarly, material ejected from the supernova must get slowed down since it is transferring momentum to the ISM gas. Since the supernova ejecta also becomes cold via expansion[1], this slowdown also occurs via a shock, known as the **reverse shock**, propagates inward in the frame of the ejected gas (but outward in the lab frame). Between the blast wave and the reverse shock lies a **contact discontinuity**, which separates the shocked supernova ejecta and shocked ISM. The contact discontinuity is not a shock: since the material on both sides moves subsonically (both have already been shock-heated), the contact discontinuity is a surface of pressure equilibrium between material of different composition and entropy.

Now, let us consider the simplest case of a spherically symmetric explosion of a star in a uniform medium with hydrogen density n_0 and temperature T_0.

1.2. *The Free-expansion Phase*

This phase is defined by the fact that the velocity of the ejecta is constant with time. Typical ejecta velocity is

$$v_{\rm ej} \approx \sqrt{\frac{2E_0}{M_{\rm ej}}} = 1.00 \times 10^4\,{\rm km\,s^{-1}} E_{51}^{1/2} \left(\frac{M_\odot}{M_{\rm ej}}\right)^{1/2} \tag{1}$$

This velocity is far greater than the sound speed in the surrounding material, and the expanding ejecta will therefore drive a fast shock into the circumstellar medium.

[1] in the sense that its sound speed is small compared to the expansion velocity since the pressure of the shocked CSM exceeds the thermal pressure in the ejecta.

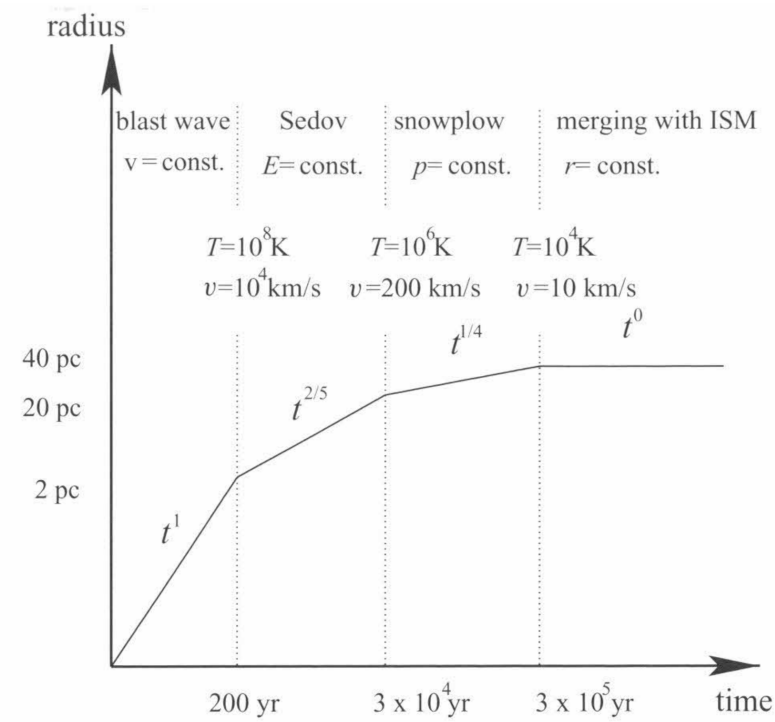

Figure 1. An illustration of the four different phases in the dynamics of a SNR. Fig 4.9 of Rosswog & Brüggen (2011).

The outward traveling shock heats the shocked ISM to very high temperatures. Thermal bremsstrahlung is seen in X-rays, and synchrotron emission is seen in radio from ejected particles spiralling around μG magnetic fields. The reverse shock has a finite amount of material to propagate inwards, shock-heats the ejecta (which was previously been cooled by adiabatic expansion), and dies away.

The condition for being in this phase is that $M_{\rm ejecta} \gtrsim M_{\rm swept}$, that is, the mass swept up by the ejecta from the ISM is not a significant fraction of the ejecta mass. Therefore, the radius of the blastwave when this ends is

$$R_1 = \left(\frac{3M_{\rm ej}}{4\pi\rho_0}\right)^{1/3} = 5.88\times10^{18}\,{\rm cm}\left(\frac{M_{\rm ej}}{M_\odot}\right)^{1/3} n_0^{-1/3} \tag{2}$$

and the time when it ends is

$$t_1 \approx \frac{R_1}{v_{\rm ej}} = 186\,{\rm yr}\left(\frac{M_{\rm ej}}{M_\odot}\right)^{5/6} E_{51}^{-1/2} n_0^{-1/3} \tag{3}$$

As is shown in Figure 1, this phase lasts about 200 years.

Examples: (a) SN 1987A is still mostly in free expansion. (b) Cas A is at the end of its free expansion phase.

1.3. *The Adiabatic Phase (also called Sedov-Taylor or Blast-Wave Phase)*

In this phase, the mass of the swept up ISM dominates the mass of the ejected material ($M_{\rm ejecta} < M_{\rm swept}$). At this stage, the blast wave is left to propagate outward through the ISM, carrying the SN's energy but leaving he ejecta behind. However, the density of the material behind the shock is not yet large enough for cooling to be significant. Therefore, the gas expands adiabatically. The blast wave is pushed forward by the thermal pressure of the ball of shock-heated gas behind it. The SNR now enters a phase that can be approximated by idealizing the problem as a "point explosion" injecting only energy E_0 into a uniform medium of density ρ_0.

Assuming that the mass in the SNR is made up by the shocked interstellar gas and $M_{\rm ej}$ can be neglected, the total energy of the expanding remnant is

$$E_{\rm total} = E_0 = \frac{4\pi}{3}R^3\rho_0(\varepsilon_{\rm int}+\varepsilon_{\rm kin}) \tag{4}$$

where the kinetic energy per unit mass of the shocked gas is

$$\varepsilon_{\rm kin} = \frac{1}{2}v_1^2 = \frac{1}{2}\left(\frac{2}{\gamma+1}v_s\right)^2 \tag{5}$$

Since $v_{\rm ej} \gg c_s$, we can make use of the strong shock approximation, $u_0 = -v_s$, and the internal energy per unit mass for the shocked gas is

$$\varepsilon_{\rm int} = \frac{P_1}{(\gamma-1)\rho_1} = \frac{\left(\frac{2}{\gamma+1}\right)\rho_0 u_0^2}{(\gamma-1)\frac{\gamma+1}{\gamma-1}\rho_0} = \frac{1}{2}\left(\frac{2}{\gamma+1}v_s\right)^2 \tag{6}$$

Hence, writing v_s as $\dot{R}$, inserting Eq. 5 and 6 into Eq. 4:

$$E_0 = \frac{3\pi}{4}R^3\dot{R}^2\rho_0 \tag{7}$$

From dimensional analysis considerations, with the initial condition $R(0) = 0$, we can derive the Sedov-Taylor solution by integrating Eq (7) to get

$$\begin{aligned} R &= \left(\frac{25}{3\pi}\right)^{1/5}\left(\frac{E_0}{\rho_0}\right)^{1/5} t^{2/5} \\ &= 5.6E_{51}^{1/5}n_0^{-1/5}\left(\frac{t}{10^3\,{\rm yr}}\right)^{2/5}\ {\rm pc} \end{aligned} \tag{8}$$

The expansion velocity can be found by taking the first derivative of $R(t)$:

$$\begin{aligned} \dot{R} &= \frac{2}{5}\left(\frac{25}{3\pi}\right)^{1/5}\left(\frac{E_0}{\rho_0}\right)^{1/5} t^{-3/5} \\ &= 2200E_{51}^{1/5}n_0^{-1/5}\left(\frac{t}{10^3\,{\rm yr}}\right)^{-3/5}\ {\rm km\,s^{-1}} \end{aligned} \tag{9}$$

The amount of shocked interstellar gas at time t is given by

$$\begin{aligned} M_s &= \frac{4\pi}{3}R^3\rho_0 \\ &= 18E_{51}^{3/5}n_0^{2/5}\left(\frac{t}{10^3\,{\rm yr}}\right)^{6/5} M_\odot \end{aligned}$$

Since this is much greater than ejecta mass, the assumption that we made in Eq. (4) is justified.

In an ideal gas, each particle contributes equally to the pressure. Assuming that the remnant is totally ionized, we see that the internal energy of the shocked gas is

$$\varepsilon_{\rm int} = \frac{3}{2}\frac{kT}{\mu m_p} \tag{10}$$

Let this equal to Eq. 6, we get

$$T = \frac{2(\gamma-1)}{(\gamma+1)^2}\frac{\mu m_p}{k}u_0^2 = \frac{3}{16}\frac{\mu m_p \dot{R}^2}{k} = 10^8\mu E_{51}^{2/5}n_0^{-2/5}\left(\frac{t}{10^3\,{\rm yr}}\right)^{-6/5}\ {\rm K} \tag{11}$$

When $t = 10^4$ yr, $\mu = 0.6$, $E_{51} = 1$, $n_0 = 1$, $T = 3.8 \times 10^6$ K. The typical temperature at this phase is 10^6–10^7 K, and the SNR strongly emits X-rays, mainly through thermal bremsstrahlung. When $T < 10^6$ K, some ions start to recombine to form atoms, which show emission lines and lead to strong cooling of the gas. The Sedov-Taylor phase is terminated when radiative losses behind the outward moving blast wave start to become important and a thin shell of cooled gas forms near this outer boundary.

Examples: (a) Tycho's supernova. (b) SN1054 (which gave birth to the Crab Nebula) should have a radius ∼5 pc and expansion velocity $\sim 2000\,{\rm km\,s^{-1}}$, if it is still in the blastwave phase of expansion. However, the Crab Nebula is observed to have a radius of 3 pc and an expansion velocity of $900\,{\rm km\,s^{-1}}$. In fact, the Crab is not exactly spherical and its expansion is not exactly energy-conserving, so the discrepancy is not surprising.

1.4. *The Radiative Phase (or Snowplow Phase)*

The shock wave continues to cool, and once temperatures drop below about $\sim 5 \times 10^5$ K, electrons start recombining to form heavier elements (in particular, oxygen line emission, see Figure 2). This recombination process radiates energy much more efficiently than the thermal X-rays and synchrotron emission produced thus far, and causes the temperature immediately behind the shock frount to drop even further. Pressured by the hot gas, the cooled gas will be compressed, forming a high-density shell. With efficient radiative cooling, the expansion can no longer be considered adiabatic. The euqation of motion will eventually be governed by the conservation of momentum as the ejecta undergoes a completely inelastic collision with the surrounding ISM.

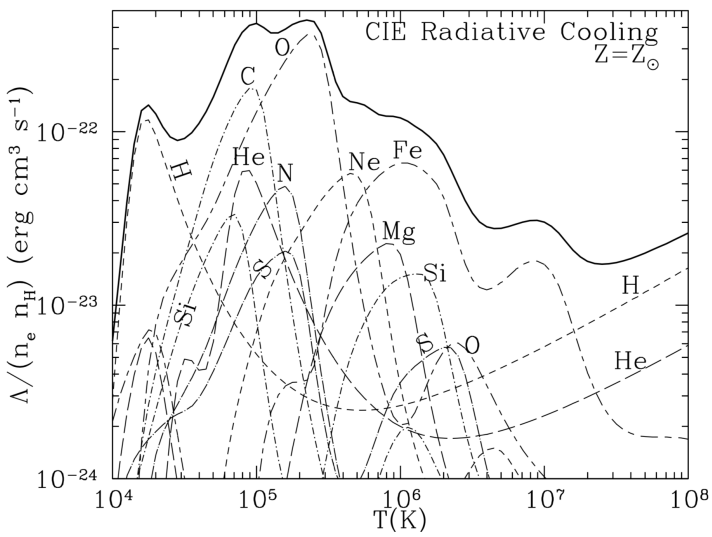

Figure 2. Radiative cooling function. Fig 34.3 of Draine (2011).

Assuming that the momentum-conserving phase begins at $t = t_0$ and the momentum of the shell, $\mathcal{P}_0 = M_s v$, remains constant hereafter with the value of

$$\mathcal{P}_0 = \left(\frac{4\pi}{3} R_0^3 \rho_0\right) \dot{R}_0$$

We have the equation of motion after t_0:

$$\mathcal{P}_0 = \left(\frac{4\pi}{3} R^3 \rho_0\right) \dot{R} \tag{12}$$

This can be integrated with respect to time to give

$$\frac{1}{4} R^4 = \frac{3\mathcal{P}_0}{4\pi\rho_0} t + C$$

where

$$C = \frac{1}{4} R_0^4 - R_0^3 \dot{R}_0 t_0$$

The solutions to the equation of motion are

$$R = R_0 \left[1 + \frac{4\dot{R}_0 (t - t_0)}{R_0}\right]^{1/4} \tag{13}$$

$$\dot{R} = \dot{R}_0 \left[1 + \frac{4\dot{R}_0 (t - t_0)}{R_0}\right]^{-3/4} \tag{14}$$

For $t \gg t_0 + R_0/\dot{R}_0$:

$$R \propto t^{1/4}$$
$$\dot{R} \propto t^{-3/4}$$

The shell propagates with constant radial momentum and piles up the ambient material like a *snowplow*. Towards the end of this phase, the expansion velocity becomes sonic or subsonic with respect to the interstellar sonic or magnetosonic speed, and the remaining energy of the ancient supernova remnant is finally dissipated through turbulent cascade.

1.5. *Merger with the ISM*

Occurs when $v_s = c_s$ is the sound speed.

2. STELLAR WIND

2.1. *Wind Velocity and Mass Loss Rate*

All hot stars, be they either young and massive or old low-mass objects at the cores of planetary nebulae, produce fast winds driven by **radiation pressure**. Radiation from the central star is scattered by the atmosphere, so that the momentum flux in the radiation field, L_*/c, is transferred to the atmospheric gas, to produce an outflowing wind. If v_w is the terminal velocity attained by the wind, and $\dot{M}_w$ its mass flux:

$$\dot{M}_w v_w = \frac{\eta L_*}{c} \tag{15}$$

Here, the factor η accounts for the fact that each photon may be scattered many times before it escapes, enhancing the total amount of momentum that can be deposited.

This process is limited, since the wind cannot carry away more energy than is produced by the star:

$$\frac{1}{2}\dot{M}_w v_w^2 < L_* \tag{16}$$

In stars, the full application of the radiatively driven wind theory shows that $\eta \sim 3$–4. In such radiatively driven winds, the outflow velocity is a factor $1 \lesssim \epsilon \lesssim 3$ times the escape velocity at the base of the outflow

$$v_w = \epsilon \left(\frac{GM_*}{r_*}\right)^{1/2} \sim 437\epsilon \left(\frac{M_*/M_\odot}{R_*/R_\odot}\right)^{1/2} \mathrm{km\,s^{-1}} \tag{17}$$

A massive O-star will spend ∼90% of its life on the main sequence, during which it blows a fast, tenuous wind with velocity $v_w \approx 2000\,\mathrm{km\,s^{-1}}$, and mass loss rate $\dot{M} \approx 10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$. In the subsequent red supergiant (RSG) phase the mass loss rate increases to $\dot{M} \approx 10^{-5}\,M_\odot\,\mathrm{yr}^{-1}$ but wind velocity decreases to $\approx 20\,\mathrm{km\,s^{-1}}$. This can be understood using Eq. (17) since RSGs are very puffy.

Most massive stars will explode in the RSG phase, but the highest mass stars may enter the Wolf-Rayet star phase. In this phase the stars have lost all of their hydrogen envelopes (resulting in a decrease in radius). Their mass loss rates are high, $\dot{M} \approx 10^{-5}\,M_\odot\,\mathrm{yr}^{-1}$, with a high wind velocity $v_w \approx 1000\,\mathrm{km\,s^{-1}}$.

2.2. *The Evolution of a (Fast) Wind Bubble*

For OB and WR stars, $v_w \gg c_s$, the wind will initially drive a shock into the H II region – the outer shock in Figure 3 – with radius R_s and velocity v_s. The shocked ambient medium accumulates in a shell behind the outer shock front. When the dynamical pressure of the wind drops to the thermal pressure of the surrounding gas, a second shock front will form to slow the stellar wind down (analogous to the reverse shock in the early supernova evolution). A contact discontinuity separates the shocked ambient medium from the shocked wind material. If there were no instabilities, and diffusion can be neglected, these two regions would not mix. The innermost region contains the unshocked stellar wind.

The evolution of stellar winds can be understood similarly as in our previous discussion of SNRs: (i) free-expansion phase (ii) adiabatic phase (iii) snowplow phase.

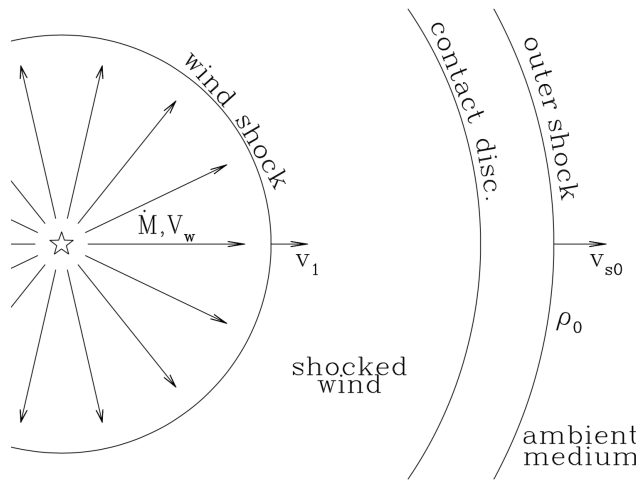

Figure 3. Structure of the bubble produced by a fast stellar wind (not to scale). Fig 38.1 of Draine (2011).

(i) At very early times, the wind undergoes essentially free expansion, with $R_s \approx v_w t$ until the wind mass and swept-up mass are comparable: $\dot{M}t \approx (4\pi/3)\rho_0(v_w t)^3$. The free expansion ends at a time

$$t_1 \approx \left(\frac{3\dot{M}}{4\pi\rho_0 v_w^3}\right)^{1/2} = 80\, n^{-1/2}\dot{M}_{-6}^{1/2} v_{w,8}^{-3/2}\,\mathrm{yr} \tag{18}$$

where $v_w \equiv 10^3 v_{w,8}\,\mathrm{km\,s^{-1}}$.

(ii) In shocked material (wind and ISM), kinetic energy of the wind is deposited in this region in the form of heat, because this gas has goen through a strong shock. This crates a temperature of $\sim 10^7\,\mathrm{K}$. There is no significant density enhancement at the contact surface since there is negligible energy loss by radiative cooling.

(iii) Although radiative cooling is not yet important, the temperature of the bubble nevertheless will decrease with time because work is done in increasing the bubble volume. When the swepted-up ISM gas has cooled adiabatically to a tmeperature of $10^6\,\mathrm{K}$, line emission becomes the dominant cooling process and the temperature of the shell rapidly decreases to $10^4\,\mathrm{K}$.

We see from Eq. (11) that a change in temperature from $10^7\,\mathrm{K}$ to $10^6\,\mathrm{K}$ corresponds to a change in jump velocity by a factor of $\sqrt{10}$. From Eq. (9) we see that in an adiabatic phase $v_s \propto t^{-3/5}$. Thus, denote duration of phase (ii) as t_2, we have $(t_2/t_1)^{3/5} \sim \sqrt{10}$, which gives $t_2 \sim 6.8t_1 \approx 550\,\mathrm{yr}$. Both are such a short time that although important for the dynamical developement of the wind bubble, the free expansion and adiabatic expansion phase are unlikely to be observed. The snowplow phase persists for as long as the star is able to sustain a powerful wind, which corresponds to few Myr for massive stars.

2.3. *Interacting Stellar Winds*

In constrast to the interstellar bubble case where a stellar wind runs into a stationary, constant density medium, the interacting winds process considers the possibility that both media have a density and velocity structure. This mechanism was first proposed to explain the origin of planetary nebulae and has since been applied to a variety of other astrophysical phenomena (e.g., ring nebulae around massive stars, supernovae, young star outflows, relativistic jets in AGN).

We consider collision of two flows:

- AGB phase wind ($v_w \sim 10\,\mathrm{km\,s^{-1}}$, $\dot{M} \sim 10^{-4}\,M_\odot\,\mathrm{yr^{-1}}$).
- Central Star of planetary nebula (CSPN) phase wind ($v_w \sim 2000\,\mathrm{km\,s^{-1}}$, $\dot{M} \sim 10^{-7}\,M_\odot\,\mathrm{yr^{-1}}$).

The interaction of the fast winder with the pre-existing slower sind produces a shell containing both AGB ejecta and shocked fast wind material. At the same time that this shell accumulation is occuring, the ultraviolet radiation from the increasingly host central star will photoionize the shell. The balance between the photoionization and recombination rates in the material leads to the prominent Hα emission of planetary nebulae. The density also tend to be low enough that the forbideen line radiation such as the [O III] λ5007 line is prominent.

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